

Ratna Vamsi

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Summary

Data Scientist with over 3.2 years of expertise in machine learning, NLP, deep learning, and computer vision, specializing in transforming complex data into actionable insights. Currently expanding knowledge in MLOps and advanced Generative AI (GenAI), with a focus on building end-to-end AI solutions for document processing and automation. Proficient in prompt engineering, LLMs, and cloud deployments, with a commitment to driving operational efficiencies through data-driven innovations.

Experience

- Flydocs India** 7-2022 - 8-2024
Data scientist
Spearheaded ML solutions for automated document processing using OCR, YOLO, and NLP models, achieving a substantial increase in document analysis efficiency.
Constructed NLP pipelines for entity extraction, document classification, and relationship mapping using BERT and spaCy, enhancing data insight generation.
Applied MLOps best practices with MLflow for tracking and model management, facilitating smooth deployment and scalability through Azure ML Studio and AWS SageMaker.
Pioneered hyperparameter tuning and prompt engineering techniques, significantly improving model accuracy and operational workflows.
- Zabda Technology** 2-2022 - 6-2022
Data scientist
Developed predictive analytics models and deep learning solutions for complex NLP applications, leveraging TensorFlow and Keras to meet business needs.
Conducted data preprocessing and feature engineering to optimize model performance, resulting in accurate and efficient model outputs.
Collaborated closely with stakeholders, transforming business requirements into data-driven solutions and actionable insights.
- Bepec solutions** 4-2021 - 10-2021
Data scientist internship
Executed comprehensive data analysis and built predictive models, enhancing decision-making capabilities through data insights.
Focused on data quality by leading data cleaning and preprocessing efforts, ensuring reliable datasets for modeling.
Documented workflows and methodologies, ensuring transparency and enabling smooth knowledge transfer.

Education

- Sabarmati University** 2018
Bachelor of computer science
65%

Skills

- Data Science & Machine Learning : Python, TensorFlow, Keras, BERT, Scikit-Learn, spaCy, NLTK, Pandas, NumPy
- Natural Language Processing (NLP): Named Entity Recognition, Sentiment Analysis, Text Classification
- Computer Vision: OpenCV, YOLO, Tesseract OCR, Azure Form Recognizer, pdf2image, Image Annotation (CVAT)
- MLOps & Cloud: MLflow, Docker, Kubernetes, Azure ML Studio, AWS SageMaker, GitLab, CI/CD
- Generative AI: GenAI, LLMs, LangChain, Hugging Face, OpenAI, prompt engineering, fine-tuning, BLIP
- Big Data & Data Processing: SQL, PySpark, Data Lakes (Azure Blob Storage)
- DevOps & Deployment: FastAPI, Flask, CI/CD pipelines, Dockerized deployments, Microservices, RESTful APIs, Version Control (Git, GitHub)
- Experiment Tracking & Model Management: MLflow (for experiment tracking, model versioning, lifecycle management), DVC (Data Version Control) . knowledge
- Data Visualization & BI: MS SQL, PowerBI, Matplotlib, Seaborn
- Additional Tools: Jupyter Notebook, VS Code, Anaconda, Google Colab, Regex, Fuzzy Logic

Projects

- **Automated Data Retrieval and Transformation in Azure Blob Storage**

Retrieve HTML data and Inventory Excel data from Azure Blob storage - upload Container. Get Delivery Document Abbreviations, Work package folder abbreviations, Folder name abbreviations and Filename abbreviations from Azure Blob Storage - staticfiles Container. Remove punctuations & special characters and replace strings use regex. An excel is generated with list of old filenames, new filenames and levenshtein distance between old and new filenames

Technology:- python , pandas, fuzzy logic, regex, azure blob service

- **Image Processing with OCR and Azure Services**

OCR, or Optical Character Recognition, is a process of recognizing text inside images and converting it into an Excel. These images could be of handwritten text, printed text like receipts and set covers images. We are using azure form recogniser API's. We want only important features from these images. For each image to be fed to Document Reader API service of Azure, we need image url and the image url has a SAS token generated by Azure along with its filename. Hence, for the extracted blob name, generate a SAS token using generate_blob_shared_access method. Stitch the blob name and SAS url to the specified path and construct the image url. Provide the image url to Document Reader API and the results are extracted as json. From the json file, key value pairs are extracted and stored in a dictionary.

Technology :- python , pandas, fuzzy logic, azure blob service , azure form recogniser

- **Document Processing Automation**

Converted PDF files into images using pdf2image library, facilitated manual annotation with CVAT, and implemented YOLOv8 for efficient object detection. Developed a Python automation script leveraging OpenCV and OCR, showcasing a comprehensive approach to document analysis and processing. Resulted in an impactful solution for enhanced efficiency in handling document-based data.

Technology :- python, pandas, pdf2image, CVAT , Yolo8 , OCR , opencv , Bert

- **Onboarding Acceleration with Document Processing***

Implementing a cutting-edge Document-based Question Answering solution. This solution, powered by advanced models like deepset/roberta-base-squad2 and layoutlm-document-qa, efficiently extracted redelivery and acceptance details from complex aircraft lease documents. The project significantly reduced onboarding timelines, streamlined asset creation, and navigated the intricacies of leased aircraft contracts. Leveraging technologies such as Azure OCR and Azure Blob Storage, the solution exhibited adaptability to various document formats and contributed to an overall optimization of onboarding workflow.

Technology :- Bert, Azure , OCR, opencv, python